

Theory of Flow in a Confined Two Aquifer System

SHLOMO P. NEUMAN AND PAUL A. WITHERSPOON

University of California, Berkeley, California 94720

Abstract. A complete analytic solution has been developed for the problem of flow to a well in a confined infinite radial system composed of two aquifers that are separated by an aquitard. The well, which is represented by a line sink, completely penetrates only one of the aquifers and discharges at constant rate. The present solution differs from previous work in that the effects of storage in the aquitard and drawdown in the unpumped aquifer are both considered. Asymptotic solutions for small values of time are also included. The analytic solutions have been evaluated numerically for selected cases. The validity of these solutions has been verified independently using the finite element method of analysis.

NOTATIONS

a ,	parameter of Hankel transform;
b_i ,	thickness of i th aquifer, L ;
b_j' ,	thickness of j th aquitard, L ;
$\text{erfc}(x)$,	complementary error function of x ;
$I_0(x)$,	zero order modified Bessel function of the first kind;
$J_0(x)$,	zero order Bessel function of the first kind;
$K_0(x)$,	zero order modified Bessel function of the second kind;
K_i ,	permeability of i th aquifer, L/T ;
K_j' ,	permeability of j th aquitard, L/T ;
p ,	parameter of Laplace transform;
Q_i ,	pumping rate from i th aquifer, L^3/T ;
r ,	radial distance from pumping well, L ;
r/B_{ii} ,	$r\sqrt{K_i'/K_i b_i b_j'}$;
$S_{s,i}$,	specific storage of i th aquifer, L^{-1} ;
$S_{s,j}'$,	specific storage of j th aquitard, L^{-1} ;
s_i ,	drawdown in i th aquifer, L ;
s_j' ,	drawdown in j th aquitard, L ;
$\bar{s}$,	Laplace transform of s ;
$\bar{s}_i$,	Hankel transform of $\bar{s}$;
s_D ,	dimensionless drawdown ($4\pi T_i s/Q_i$);
T_i ,	transmissibility of i th aquifer ($K_i b_i$), L^2/T ;
t ,	pumping time, T ;
$t_{D,i}$,	dimensionless time for pumped (i th) aquifer ($K_i t/S_{s,i} r^2$);
$t_{D,j}'$,	dimensionless time for j th aquitard ($K_j' t/S_{s,j}' r^2$);
z ,	vertical distance above bottom of aquitard 1, L ;
α_i ,	$K_i/S_{s,i}$, $L^2 T^{-1}$;
α_j' ,	$K_j'/S_{s,j}'$, $L^2 T^{-1}$;
β_{ij} ,	$r/4b_i\sqrt{K_j'S_{s,j}'/K_i S_{s,i}}$.

INTRODUCTION

In dealing with transient or nonsteady flow of ground-water to wells, the traditional approach has been to focus attention on effects in the

particular aquifer from which fluid is being withdrawn. If the aquifer is confined above and below by effectively impermeable strata, methods of analysis based on the well-known Theis [1935] solution should be applicable.

Quite often, however, the aquifer from which water is being withdrawn is merely part of a complex multiple aquifer system. Such a system consists of a series of aquifers separated from each other by confining beds of relatively low permeability. When the flow of water through these confining beds is sufficient to be detected by observations of drawdown in the aquifer being pumped, the confining beds are called 'aquitards' and the aquifer is referred to as being 'leaky'. When such flow cannot be detected in the aquifer, the confining beds are called 'aquicludes' and the aquifer is termed 'slightly leaky' [Neuman and Witherspoon, 1968].

The fact that aquifers are part of a more complex geohydrologic system has long been recognized. DeGlee [1930] was apparently the first person to formulate a mathematical approach for the steady state leakage of water through less permeable layers into an aquifer that is being pumped. This was followed by several more steady state solutions by others for related problems. Jacob [1946] extended the treatment to include the transient effects of leakage, and this also has been followed by a series of solutions for particular problems. Later a modified approach to this problem was presented by Hantush [1960]. A more complete review of

pertinent literature is given elsewhere [Neuman and Witherspoon, 1969a, 1969b].

Most of this work has focused attention on the effects within the aquifer being pumped. However, the difficulty with this approach is that observations within the pumped aquifer alone may not be adequate to characterize the hydrologic properties of the aquifer and its associated confining beds. Indeed as we attempt to demonstrate in another paper [Neuman and Witherspoon, 1969a], analyses based on current theories of leaky aquifers can sometimes lead to gross errors.

The withdrawal of water from one layer of a multiple aquifer system can and frequently does have measurable effects in layers other than the pumped aquifer itself. The steady state interaction between aquifers was apparently first analyzed by Myatiev [1946, 1947]. His work has been summarized by Polubarinova-Kochina [1962, p. 385]. Nonsteady state interaction between two aquifers has recently been studied by Hantush [1967]. The importance of observing drawdowns in an aquiclude overlying an aquifer in which the storage of natural gas is contemplated was pointed out by Witherspoon *et al.* [1962]. Neuman and Witherspoon [1968] have developed analytic solutions for the case of a slightly leaky aquifer, and examples of the application of these solutions to field data have also been developed [Witherspoon, *et al.*, 1967; Witherspoon and Neuman, 1967].

Thus far investigators concerned with nonsteady flow in complex systems have either neglected storage in the confining beds, or ignored drawdown in the unpumped aquifers, or both. The assumption of no storage in the confining beds implies that at any instant of time drawdown varies linearly across such beds, which is certainly not the case at small values of time. At large values of time this may eventually be realized if the system approaches a quasi-steady state.

The second assumption of no drawdown in the unpumped aquifer is valid at sufficiently small values of time, in which case the effects of this aquifer on the rest of the system can be ignored. However, as will be shown below, these effects can be quite significant at large values of time. Therefore proper consideration must be given to both storage and drawdown

at all points if one wishes to obtain a complete understanding of the nature of flow in a multiple aquifer system.

As a first step toward the analysis of this problem we shall present an analytic approach to transient flow in a confined two aquifer system. We shall make only one assumption that the direction of flow is vertical in the aquitard (or aquiclude) and horizontal in the aquifers. The validity of this assumption has been investigated using the finite element method, a new numeric technique that has recently been described by Javandel and Witherspoon [1968, 1969]. It was found that when the permeabilities of the aquifers are two or more orders of magnitude greater than that of the aquitard, the errors introduced by this assumption are usually less than 5%. More details can be found elsewhere [Neuman and Witherspoon, 1969b].

Analytic solutions for the case of a system composed of three aquifers and two aquitards have also been developed [Neuman and Witherspoon, 1969b]. The equations have not yet been evaluated and will not be presented here.

THEORETIC DEVELOPMENT

Consider a confined system of two aquifers that enclose an aquitard as shown schematically in Figure 1. A well of infinitesimal radius is completed in the lower aquifer and discharges at constant rate Q_1 . Each layer is homogeneous, isotropic, horizontal, and of infinite radial extent. The system remains saturated at all times, and Darcy's law applies.

As mentioned earlier we shall assume that the direction of flow is vertical in the aquitard and horizontal in both aquifers. One apparent

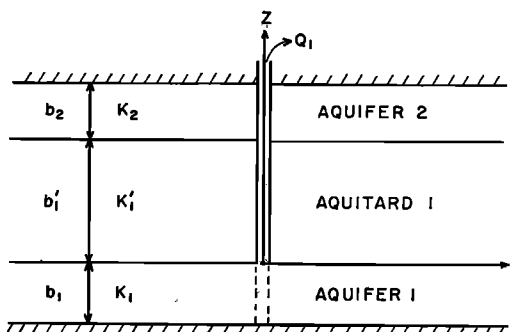


Fig. 1. Schematic diagram of two aquifer system.

implication of this assumption is that the unpumped aquifer must be thin enough so that drawdown across its thickness at any time may be regarded as constant. The principle here is analogous to that of temperature distribution in a cooling fin [Carslaw and Jaeger, 1959, p. 141]. From a practical standpoint, however, we have found that the thickness of the unpumped aquifer has little effect on the applicability of our solutions [Neuman and Witherspoon, 1969b]. Similarly no restriction on the thickness of the pumped aquifer is necessary because the radial gradients induced by the well are much larger than the vertical gradients in this layer caused by inflow from the aquitard.

In a manner similar to that of Hantush [1960] we shall formulate the problem as follows:

$$\left. \begin{aligned} \frac{\partial^2 s_1}{\partial r^2} + \frac{1}{r} \frac{\partial s_1}{\partial r} + \frac{K_1'}{T_1} \frac{\partial s_1'}{\partial z} \Big|_{z=0} &= \frac{1}{\alpha_1} \frac{\partial s_1}{\partial t} & (a) \\ s_1(r, 0) &= 0 & (b) \\ s_1(\infty, t) &= 0 & (c) \\ \lim_{r \rightarrow 0} r \frac{\partial s_1}{\partial r} &= -\frac{Q_1}{2\pi T_1} & (d) \end{aligned} \right\} \text{Aquifer 1 (1)} \quad (\text{pumped})$$

$$\left. \begin{aligned} \frac{\partial^2 s_1'}{\partial z^2} &= \frac{1}{\alpha_1'} \frac{\partial s_1'}{\partial t} & (a) \\ s_1'(r, z, 0) &= 0 & (b) \\ s_1'(r, 0, t) &= s_1(r, t) & (c) \\ s_1'(r, b_1', t) &= s_2(r, t) & (d) \end{aligned} \right\} \text{Aquitard 1 (2)}$$

$$\left. \begin{aligned} \frac{\partial^2 s_2}{\partial r^2} + \frac{1}{r} \frac{\partial s_2}{\partial r} - \frac{K_1'}{T_2} \frac{\partial s_1'}{\partial z} \Big|_{z=b_1'} &= \frac{1}{\alpha_2} \frac{\partial s_2}{\partial t} & (a) \\ s_2(r, 0) &= 0 & (b) \\ s_2(\infty, t) &= 0 & (c) \\ \lim_{r \rightarrow 0} r \frac{\partial s_2}{\partial r} &= 0 & (d) \end{aligned} \right\} \text{Aquifer 2 (3)} \quad (\text{unpumped})$$

After applying Laplace and Hankel transforms to equations 1, 2, and 3 and inverting the results, one obtains the complete solution to the initial boundary value problem in the

form of three equations. The mathematics involved is outlined in Appendix A. The results can be expressed in terms of five dimensionless parameters: (a) β_{11} and r/B_{11} with reference to the pumped aquifer, (b) β_{21} and r/B_{21} with reference to the unpumped aquifer, and (c) t_{D1} the dimensionless time. In the pumped aquifer

$$s_1(r, t) = \frac{Q_1}{4\pi T_1} \int_0^\infty (1 - e^{-\nu^2 t_{D1}}) \cdot \{[1 + G(y)]J_0[\omega_1(y)] + [1 - G(y)]J_0[\omega_2(y)]\} \frac{dy}{y} \quad (4)$$

In the aquitard

$$s_1'(r, z, t) = \frac{Q_1}{4\pi T_1} \frac{2}{\pi} \sum_{n=1}^\infty \frac{1}{n} \sin \frac{n\pi z}{b_1'} \cdot \int_0^\infty \left[1 - e^{-n^2 \pi^2 t_{D1}} + \frac{e^{-n^2 \pi^2 t_{D1}} - e^{-\nu^2 t_{D1}}}{1 - y^2/(n^2 \pi^2)} \right] \cdot \left\{ \left[\frac{2(r/B_{21})^2 (-1)^n y}{F(y) \sin y} - G(y) - 1 \right] J_0[\omega_1(y)] - \left[\frac{2(r/B_{21})^2 (-1)^n y}{F(y) \sin y} - G(y) + 1 \right] \cdot J_0[\omega_2(y)] \right\} \frac{dy}{y} \quad (5)$$

In the unpumped aquifer

$$s_2(r, t) = \frac{Q_1}{4\pi T_1} \int_0^\infty (1 - e^{-\nu^2 t_{D1}}) \frac{2(r/B_{21})^2}{F(y)} \cdot \{J_0[\omega_1(y)] - J_0[\omega_2(y)]\} \frac{dy}{\sin y} \quad (6)$$

where

$$t_{D1} = t_{D1}(r/B_{11})^4 / (4\beta_{11})^2 \quad (7)$$

$$G(y) = M(y)/F(y) \quad (8)$$

$$\omega_1^2(y) = \frac{1}{2}[N(y) + F(y)] \quad (9)$$

$$\omega_2^2(y) = \frac{1}{2}[N(y) - F(y)] \quad (10)$$

$$F^2(y) = M^2(y) + \left[\frac{2(r/B_{11})(r/B_{21})y}{\sin y} \right]^2 \quad (11)$$

$$M(y) = \left[\frac{(r/B_{11})^4}{(4\beta_{11})^2} - \frac{(r/B_{21})^4}{(4\beta_{21})^2} \right] y^2 - \left[\left(\frac{r}{B_{11}} \right)^2 - \left(\frac{r}{B_{21}} \right)^2 \right] y \cot y \quad (12)$$

$$N(y) = \left[\frac{(r/B_{11})^4}{(4\beta_{11})^2} + \frac{(r/B_{21})^4}{(4\beta_{21})^2} \right] y^2 - \left[\left(\frac{r}{B_{11}} \right)^2 + \left(\frac{r}{B_{21}} \right)^2 \right] y \cot y \quad (13)$$

Here $J_0[\omega_1(y)]$ must be set to zero when $\omega_1^2(y) < 0$, and the same is true of $J_0[\omega_2(y)]$ when $\omega_2^2(y) < 0$. Since equations 1, 2, and 3 are linear, one can use the above solutions and the principle of superposition to obtain solutions for any number of pumping wells in either aquifer. It should be mentioned that the above equations could be expressed in simpler terms; we have chosen to use r/B and β because they have been extensively used in the literature.

SOLUTIONS FOR CASE WHEN AQUIFERS HAVE IDENTICAL PROPERTIES

In the special case where the hydraulic properties of the two aquifers are identical, then $r/B_{11} = r/B_{21}$ and $\beta_{11} = \beta_{21}$. As a result $G(y) = M(y) = 0$, and equations 4, 5, and 6 can be simplified, respectively, to

$$s_1(r, t) = \frac{Q_1}{4\pi T_1} \int_0^\infty (1 - e^{-y^2 t D_1}) \{J_0[\omega_1(y)] + J_0[\omega_2(y)]\} \frac{dy}{y} \quad (14)$$

$$s_1'(r, z, t) = \frac{Q_1}{4\pi T_1} \frac{2}{\pi} \sum_{n=1}^\infty \frac{1}{n} \sin \frac{n\pi z}{b_1'} \int_0^\infty \left[1 - e^{-n^2 \pi^2 t D_1} + \frac{e^{-n^2 \pi^2 t D_1} - e^{-y^2 t D_1}}{1 - y^2/(n^2 \pi^2)} \right] \cdot \{ [(-1)^n - 1] J_0[\omega_1(y)] - [(-1)^n + 1] J_0[\omega_2(y)] \} \frac{dy}{y} \quad (15)$$

$$s_2(r, t) = \frac{Q_1}{4\pi T_1} \int_0^\infty (1 - e^{-y^2 t D_1}) \{J_0[\omega_1(y)] - J_0[\omega_2(y)]\} \frac{dy}{y} \quad (16)$$

where now one must use

$$F(y) = \frac{2(r/B_{11})^2 y}{\sin y}$$

$$N(y) = \left[\frac{2(r/B_{11})^4}{(4\beta_{11})^2} \right] y^2 - 2(r/B_{11})^2 y \cot y$$

As previously, the Bessel functions must be set to zero when the squares of their arguments are negative.

SOLUTIONS FOR CASE OF NO DRAWDOWN IN UNPUMPED AQUIFER

In his modified theory of leaky aquifers Hantush has published asymptotic solutions at

small and large values of time for the pumped aquifer of a three aquifer system [Hantush, 1960] and of a two aquifer system [Hantush, 1964] where drawdowns in the unpumped aquifers remain zero. A complete solution for the two aquifer case is easily derived from equations 4 and 5 by letting $K_2 \rightarrow \infty$, which means that $r/B_{21} = \beta_{21} = 0$. As a result $G(y) = 1$ since $F(y) = M(y) = N(y)$, and (4) and (5) can be simplified, respectively, to

$$s_1(r, t) = \frac{Q_1}{4\pi T_1} 2 \int_0^\infty (1 - e^{-y^2 t D_1}) J_0[\omega_1(y)] \frac{dy}{y} \quad (17)$$

$$s_1'(r, z, t) = \frac{Q_1}{4\pi T_1} \frac{4}{\pi} \sum_{n=1}^\infty \frac{1}{n} \sin \frac{n\pi z}{b_1'} \cdot \int_0^\infty \left[e^{-n^2 \pi^2 t D_1} - 1 - \frac{e^{-n^2 \pi^2 t D_1} - e^{-y^2 t D_1}}{1 - y^2/(n^2 \pi^2)} \right] \cdot J_0[\omega_1(y)] \quad (18)$$

where now one must use

$$\omega_1^2(y) = \frac{(r/B_{11})^4}{(4\beta_{11})^2} y^2 - (r/B_{11})^2 y \cot y$$

and as before the Bessel function must be set to zero when $\omega_1^2(y) < 0$.

REDUCTION TO THEIS SOLUTION

If the permeability of the aquitard vanishes ($K_1' = 0$), equation 4 should reduce to the Theis solution. To show this we define a new variable

$$x = \frac{(r/B_{11})^2}{4\beta_{11}} y \quad (19)$$

substitute this in (4) as well as in the definitions given by (7) through (13) and then let $K_1' \rightarrow 0$. Equation 4 will reduce to

$$s_1(r, t) = \frac{Q_1}{2\pi T_1} \int_0^\infty (1 - e^{-\alpha_1 t \nu}) J_0(x) \frac{dx}{x} \quad (20)$$

where $t_{D1} = \alpha_1 t / r^2$. From tables of definite integrals [Gradshteyn and Ryzhik, 1963, p. 731] one can find

$$\int_0^\infty e^{-\alpha x^2} J_0(x) \frac{dx}{x} = \frac{1}{2} \int_0^{1/\alpha} e^{-y} \frac{dy}{y}$$

where the real part of α is nonnegative. Combining this with (20) one immediately obtains

$$s_1(r, t) = \frac{Q_1}{4\pi T_1} \int_{1/(4t_{D1})}^\infty e^{-y} \frac{dy}{y}$$

which is the Theis solution.

ASYMPTOTIC SOLUTIONS FOR SMALL VALUES OF TIME

In his modified theory of leaky aquifers Hantush [1960] has presented a solution in the pumped aquifer that applies to our problem at small values of time. His solution may be written

$$s_1(r, t) = \frac{Q_1}{4\pi T_1} \int_{1/(4t_{D1})}^\infty \exp(-y) \cdot \operatorname{erfc} \left[\frac{\beta_{11}}{\sqrt{y(4t_{D1}y - 1)}} \right] \frac{dy}{y} \quad (21)$$

where

$$t_{D1} \leq \frac{1.6\beta_{11}^2}{(r/B_{11})^4}.$$

At small values of time one can develop a corresponding solution in the aquitard (Appendix B) which may be written

$$s_1'(r, z, t) = \frac{Q_1}{4\pi T_1} \int_{1/(4t_{D1})}^\infty \frac{e^{-y}}{y} \left\{ \operatorname{erfc} \left[\frac{\beta_{11} + y(z/b_1') 4\beta_{11}/(r/B_{11})^2}{\sqrt{y(4t_{D1}y - 1)}} \right] - \operatorname{erfc} \left[\frac{\beta_{11} + y(2 - z/b_1') 4\beta_{11}/(r/B_{11})^2}{(\sqrt{y(4t_{D1}y - 1)})} \right] \right\} dy \quad (22)$$

For the particular case of a semi-infinite aquitard where $b_1' \rightarrow \infty$, equation 22 reduces to

$$s_1'(r, z, t) = \frac{Q_1}{4\pi T_1} \int_{1/(4t_{D1})}^\infty \frac{e^{-y}}{y} \cdot \operatorname{erfc} \left[\frac{\beta_{11} + y\sqrt{t_{D1}/t_{D1}'}}{\sqrt{y(4t_{D1}y - 1)}} \right] dy \quad (23)$$

which is identical with a solution previously developed by Hantush (personal communication, 1960).

EVALUATION OF ANALYTIC SOLUTIONS

To demonstrate the nature of our analytic solutions, we have evaluated equations 4, 5, and 6 as well as the asymptotic equations 21 and 22, using the Zonneveld adaptation of the Adams-Moulton numeric method of integration. Since the asymptotic solutions require much less computer time and yield results that are practically identical with those from the complete solutions, the former were generally used whenever $t_{D1} \leq 1.6\beta_{11}^2/(r/B_{11})^4$.

We shall first consider the case where the aquifers have identical properties. If we choose $\beta_{11} = \beta_{21} = 0.01$ and $r/B_{11} = r/B_{21} = 0.01$, we can evaluate (4), (5), and (6) in terms of dimensionless drawdown s_D and dimensionless time with respect to the pumped aquifer t_{D1} . The results are shown in Figure 2 as a family of five curves. Two of the curves show results for the aquifers, and the other three curves represent different elevations in the aquitard. The Theis solution is included for reference purposes. Figures 3 and 4 show similar results for the cases where the above parameters are increased to 0.1 and 1.0, respectively.

To check the validity of these analytic solutions, we used the finite element method recently described by Javandel and Witherspoon [1968, 1969]. This is a new numeric procedure that is completely general with respect to geometry of the system and flow properties of individual layers. Thus results obtained by this method provide an independent check on the analytic solutions.

The finite element results are included on Figures 2, 3, and 4, and it may be seen that there is excellent agreement between the analytic and numeric solutions. At small values of time when vertical gradients in the aquitard are large, an extremely fine network is required in order to obtain results. The network used

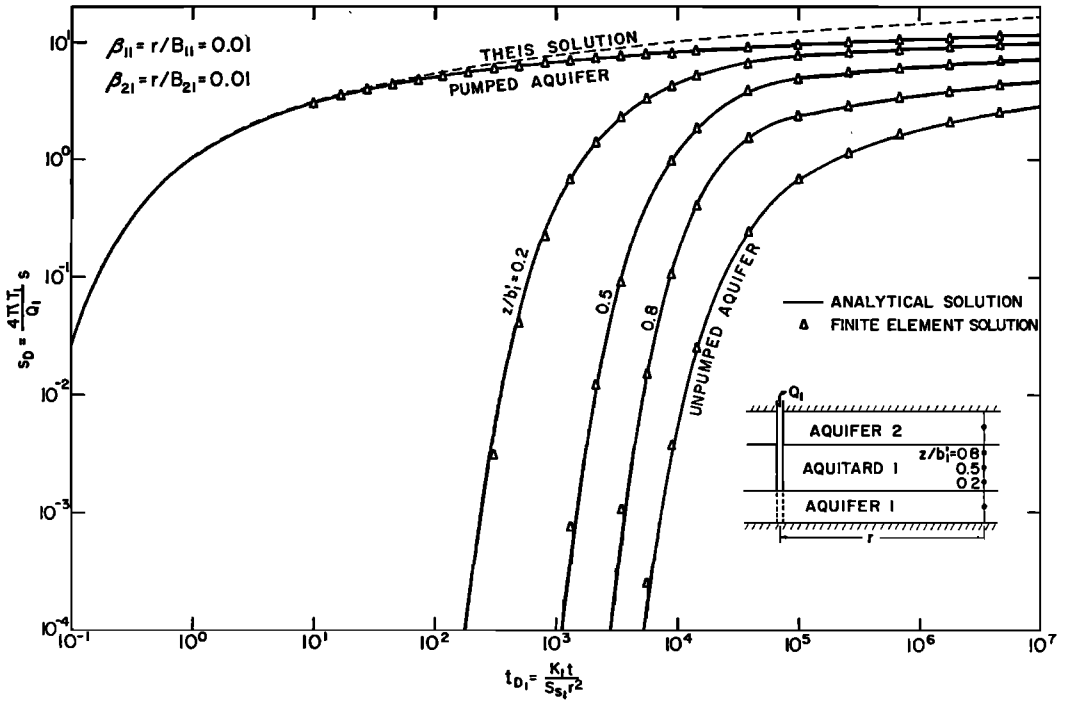


Fig. 2. Dimensionless drawdown versus dimensionless time in a two aquifer system for $\beta_{11} = \beta_{21} = r/B_{11} = r/B_{21} = 0.01$.

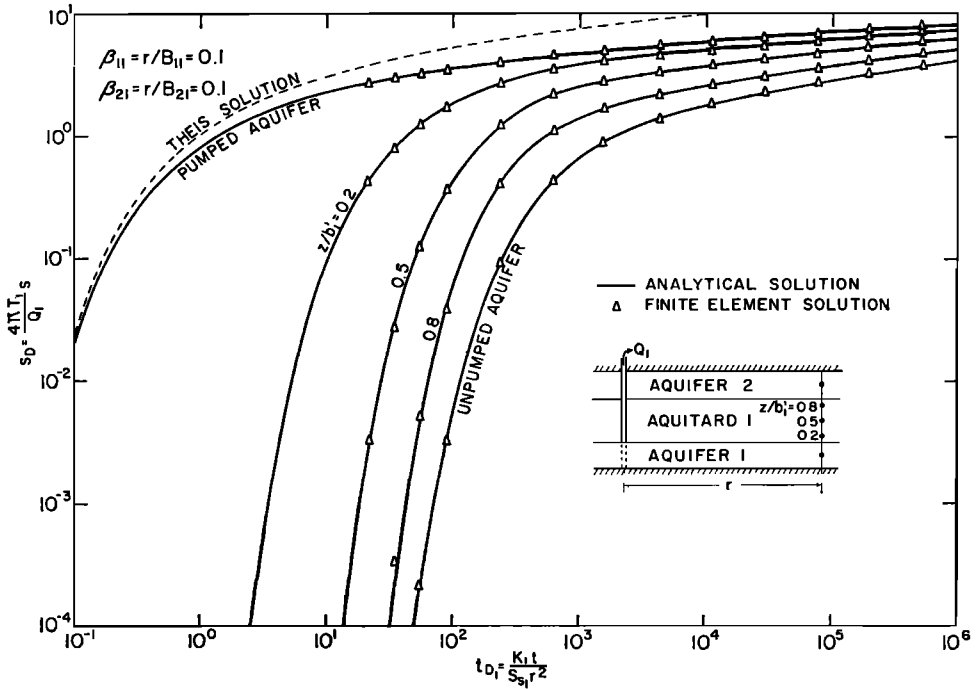


Fig. 3. Dimensionless drawdown versus dimensionless time in a two aquifer system for $\beta_{11} = \beta_{21} = r/B_{11} = r/B_{21} = 0.1$.

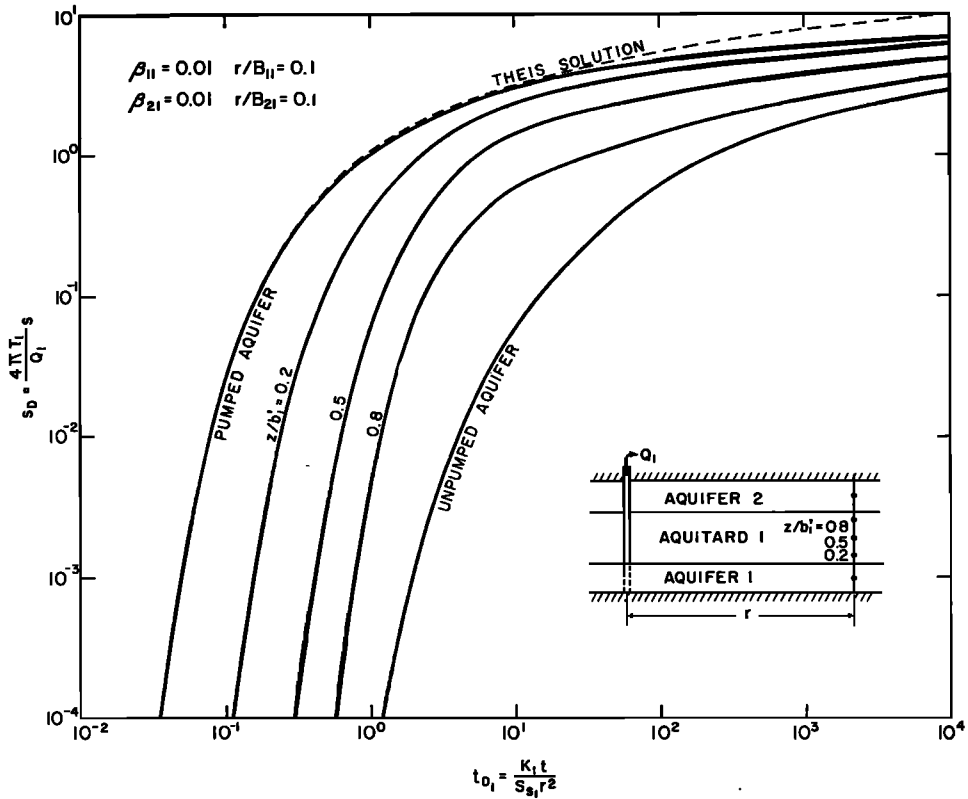


Fig. 5. Dimensionless drawdown versus dimensionless time in a two aquifer system for $\beta_{11} = \beta_{21} = 0.01$ and $r/B_{11} = r/B_{21} = 0.1$.

r/B_{21} . Now we shall consider the more general case where the properties of these aquifers are not identical. Equations 21 and 22 indicate that at $t_{D1} \leq 1.6\beta_{11}^2/(r/B_{11})^4$, the properties of the unpumped aquifer have a negligible effect on drawdowns in other parts of the system. Thus these asymptotic solutions are independent of the magnitudes of β_{21} and r/B_{21} .

At larger values of time, however, the solutions everywhere in the system will be affected by the properties of the unpumped aquifer (i.e., by β_{21} and r/B_{21}). To demonstrate this we shall consider a system where $\beta_{11} = r/B_{11} = 0.1$ and the properties of the unpumped aquifer vary in such a way that β_{21} and r/B_{21} range from zero to infinity. This corresponds to those situations where the unpumped aquifer has infinite and zero transmissibilities, respectively.

Figure 7 shows results for $z/b_1' = 0.8$ in the

aquitard and Figure 8 shows the effects in the pumped aquifer. The limiting values of t_{D1} below which the asymptotic solution (equations 21 and 22) are applicable are indicated on both figures. The results shown in both figures were obtained using the finite element method. The particular curves for $\beta_{21} = r/B_{21} = 0.1$ and 0 were also checked using the analytic solutions (equations 4 and 5). It should be mentioned that as the permeability of the unpumped aquifer approaches zero (i.e., $\beta_{21} = r/B_{21} \rightarrow \infty$) the assumption of vertical flow in the aquitard is no longer generally applicable, and consequently the analytic solutions must be used with caution.

It can be seen in both Figure 7 and 8 that when transmissibility of the unpumped aquifer becomes very large (i.e., $T_2 \rightarrow \infty$), drawdowns become constant as the pseudo steady state is reached. In this case the drawdown changes linearly across the thickness of the aquitard

[Neuman and Witherspoon, 1969b]. These results further suggest that when $T_2 > 100T_1$ (i.e., $\beta_m < 0.1 \beta_u$ and $r/B_m < 0.1 r/B_u$), one is probably justified in neglecting drawdown in the unpumped aquifer. It should be noted that the spread in the curves, which reflects the effect of the properties of the unpumped aquifer, increases significantly with vertical distance from the pumped aquifer. This is to be expected because the closer the observation point is to the unpumped aquifer, the greater effect the properties of this aquifer will have.

APPENDIX A

To solve the initial boundary value problem given by (1), (2), and (3) we first apply Laplace and then Hankel transforms to equation 2. One obtains an ordinary differential equation from (2a) and (2b). Solving this equation using (2c) and (2d), one obtains the double transform of drawdown in the aquitard

$$\begin{aligned} \bar{s}_1' &= \frac{\sinh(\sqrt{p/\alpha_1'} z)}{\sinh(\sqrt{p/\alpha_1'} b_1')} \bar{s}_2 \\ &+ \frac{\sinh[\sqrt{p/\alpha_1'}(b_1' - z)]}{\sinh(\sqrt{p/\alpha_1'} b_1')} \bar{s}_1 \end{aligned} \quad (A1)$$

Using a particular Hankel transform given by equation C3 (Appendix C) with (1d) and (3d), the same procedure can be applied to equations 1 and 3. The results for each of these equations may be written

$$\begin{aligned} -a^2 \bar{s}_1 + \frac{Q_1}{2\pi T_1 p} - \frac{p}{\alpha_1} \bar{s}_1 + \frac{K_1'}{T_1} \frac{\partial \bar{s}_1'}{\partial z} \Big|_{z=0} &= 0 \\ -a^2 \bar{s}_2 - \frac{p}{\alpha_2} \bar{s}_2 - \frac{K_1'}{T_2} \frac{\partial \bar{s}_1'}{\partial z} \Big|_{z=b_1'} &= 0 \end{aligned} \quad (A2)$$

After (A1) is differentiated and substituted in (A2), one can solve for the double transforms

$$\bar{s}_1 = \frac{C(a^2 + A_2)}{(a^2 + A_1)(a^2 + A_2) - B_1 B_2} \quad (A3)$$

$$\bar{s}_2 = \frac{CB_2}{(a^2 + A_1)(a^2 + A_2) - B_1 B_2} \quad (A4)$$

where

$$\left. \begin{aligned} C(p) &= Q_1/2\pi T_1 p \\ A_1(p) &= \frac{p}{\alpha_1} + \frac{K_1' \sqrt{p/\alpha_1'}}{T_1} \cdot \coth(\sqrt{p/\alpha_1'} b_1') \\ A_2(p) &= \frac{p}{\alpha_2} + \frac{K_1' \sqrt{p/\alpha_1'}}{T_2} \cdot \coth(\sqrt{p/\alpha_1'} b_1') \\ B_1(p) &= \frac{K_1' \sqrt{p/\alpha_1'}}{T_1 \sinh(\sqrt{p/\alpha_1'} b_1')} \\ B_2(p) &= \frac{K_1' \sqrt{p/\alpha_1'}}{T_2 \sinh(\sqrt{p/\alpha_1'} b_1')} \end{aligned} \right\} \quad (A5)$$

In the process of inverting the Hankel transforms it is convenient to write (A3) and (A4) as

$$\bar{s}_1 = \frac{C(a^2 + A_2)}{D} \left[\frac{1}{a^2 + \xi_1^2} - \frac{1}{a^2 + \xi_2^2} \right] \quad (A6)$$

$$\bar{s}_2 = \frac{CB_2}{D} \left[\frac{1}{a^2 + \xi_1^2} - \frac{1}{a^2 + \xi_2^2} \right] \quad (A7)$$

where

$$\left. \begin{aligned} D^2(p) &= 4B_1(p)B_2(p) + [A_1(p) - A_2(p)]^2 \\ \xi_1^2(p) &= \frac{1}{2}[A_1(p) + A_2(p) - D(p)] \\ \xi_2^2(p) &= \frac{1}{2}[A_1(p) + A_2(p) + D(p)] \end{aligned} \right\} \quad (A8)$$

With the aid of equation C4 the inverse Hankel transforms of (A6) and (A7) become, respectively,

$$\begin{aligned} \bar{s}_1 &= \frac{C(A_2 - \xi_1^2)}{D} K_0(\xi_1 r) \\ &- \frac{C(A_2 - \xi_2^2)}{D} K_0(\xi_2 r) \end{aligned} \quad (A9)$$

$$\bar{s}_2 = \frac{CB_2}{D} [K_0(\xi_1 r) - K_0(\xi_2 r)] \quad (A10)$$

The inverse Hankel transform of (A1) is simply

$$\begin{aligned} \bar{s}_1' &= \frac{\sinh(\sqrt{p/\alpha_1'} z)}{\sinh(\sqrt{p/\alpha_1'} b_1')} \bar{s}_2 \\ &+ \frac{\sinh[\sqrt{p/\alpha_1'}(b_1' - z)]}{\sinh(\sqrt{p/\alpha_1'} b_1')} \bar{s}_1 \end{aligned} \quad (A11)$$

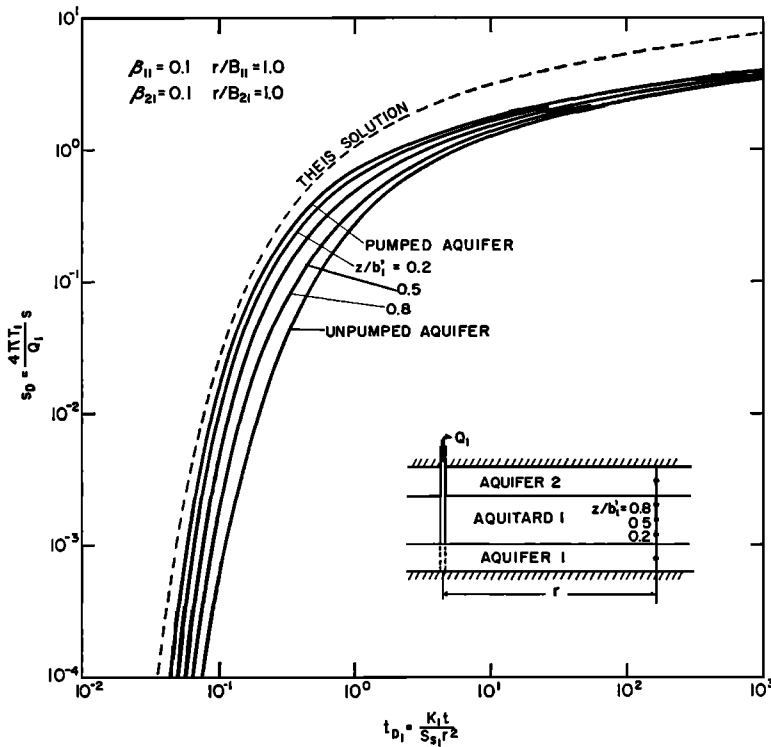


Fig. 6. Dimensionless drawdown versus dimensionless time in a two aquifer system for $\beta_{11} = \beta_{21} = 0.1$ and $r/B_{11} = r/B_{21} = 1.0$.

The inversion of the Laplace transforms of (A9) and (A10) is accomplished using Mellin's inversion formula written as

$$s(r, t) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} (e^{\lambda t} - 1) \bar{s}(r, \lambda) d\lambda \quad (\text{A12})$$

where γ is a real positive constant and λ is a complex variable replacing p .

The first step is to determine the singularities of $\bar{s}_1$ and $\bar{s}_2$ in the complex domain of λ . We note that $K_0(\xi r)$ has branch points at those values

one can obtain

$$u^2[(cd + u^2) \sinh u + u(c + d) \cosh u] = 0 \quad (\text{A13})$$

where c and d are positive real constants. For the case when $u = 0$ one can verify that $\lambda = 0$ is a root of $\xi_1^2 = 0$ but not of $\xi_2^2 = 0$. For the case when $u \neq 0$ the term inside the brackets of (A13) must vanish. Solving for e^{2u} and taking the square of the modulus, one obtains

$$e^{4x} = \frac{[x^2 - y^2 + cd - x(c + d)]^2 + y^2[c + d - 2x]^2}{[x^2 - y^2 + cd + x(c + d)]^2 + y^2[c + d + 2x]^2}$$

of λ which are roots of $\xi = 0$. By letting $\xi_1^2 = \xi_2^2 = 0$, (A8) will lead to

$$A_1 A_2 - B_1 B_2 = 0$$

Multiplying this by $(\alpha_1 \alpha_2 b_1'^4 / \alpha_1'^2) \sinh(\sqrt{\lambda / \alpha_1'} b_1')$ and introducing the complex variable

$$u = x + iy = \sqrt{\lambda / \alpha_1'} b_1'$$

This equation is only satisfied at $x = 0$. Since there is an infinite number of solutions $u = iy$ for (A13), the implication is that $K_0(\xi_1 r)$ and $K_0(\xi_2 r)$ have at most an infinite number of distinct real and nonpositive branch points of the form

$$\lambda_n = -y_n^2 \alpha_1' / b_1'^2 \quad (n = 1, 2, \dots)$$

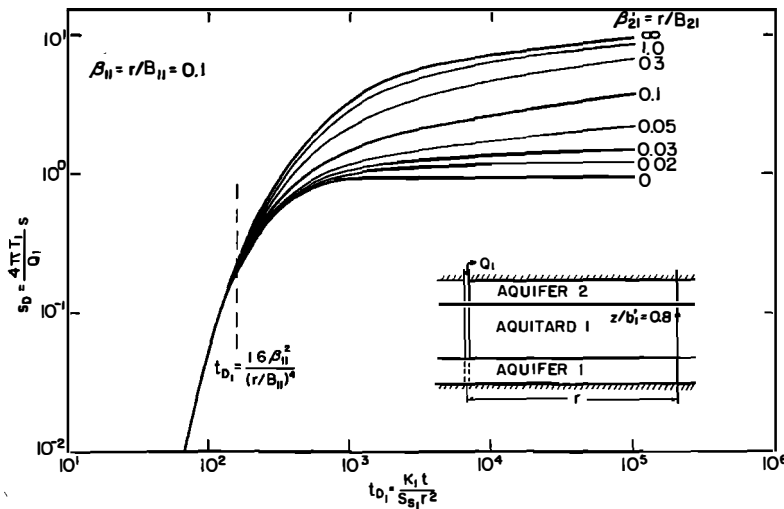


Fig. 7. Effect of unpumped aquifer on dimensionless drawdown in aquitard at $z/b_1' = 0.8$.

In a similar manner it can be shown that $D = 0$ in (A9) and (A10) does not have any roots along the real axis of the λ plane; if it has other roots, these cannot be singular points of $\bar{s}_1$ and $\bar{s}_2$ [Neuman and Witherspoon, 1969b]. In addition the roots of $\sinh(\sqrt{\lambda/\alpha_1'} b_1')$ are removable singularities of $\bar{s}_1$ and $\bar{s}_2$ and may therefore be disregarded. Thus the only singularities in addition to λ_n are simple poles at $\lambda = 0$ due to the form of C as given in (A5).

Let us consider the first part of (A9) and omit the constant $Q_1/2\pi T_1$. The path of integration will be as shown in Figure 9. Here Γ_n and Γ_n' are semicircles of radius ϵ about the roots of $\xi_1(\lambda) = 0$. According to Cauchy's theorem the integral along the entire contour vanishes, and therefore

one can write

$$\int_{\gamma-i\infty}^{\gamma+i\infty} (e^{\lambda t} - 1) \frac{A_2 - \xi_1^2}{D} K_0(\xi_1 r) \frac{d\lambda}{\lambda} \\ = -\lim_{\substack{\epsilon \rightarrow 0 \\ R \rightarrow \infty}} \left[I_{AC} + I_{CD} + I_{DD'} + I_{D'C'} \right. \\ \left. + I_{C'A'} + \sum_{n=1}^{\infty} (I_{\Gamma_n} + I_{\Gamma_n'}) \right] \quad (A14)$$

where I_{AC} and I_{CD} represent integrals for different portions of the contour. It can be shown that the integrals I_{AC} and $I_{C'A'}$ vanish as $R \rightarrow \infty$, and likewise $I_{DD'}$, I_{Γ_n} , and $I_{\Gamma_n'}$ vanish as $\epsilon \rightarrow 0$ [Neuman and Witherspoon, 1969b]. To evaluate I_{CD} , let $\lambda = (y^2 \alpha_1' / b_1'^2) e^{i\pi}$ and $\xi_1^* = \xi_1(y^2 \alpha_1' /$

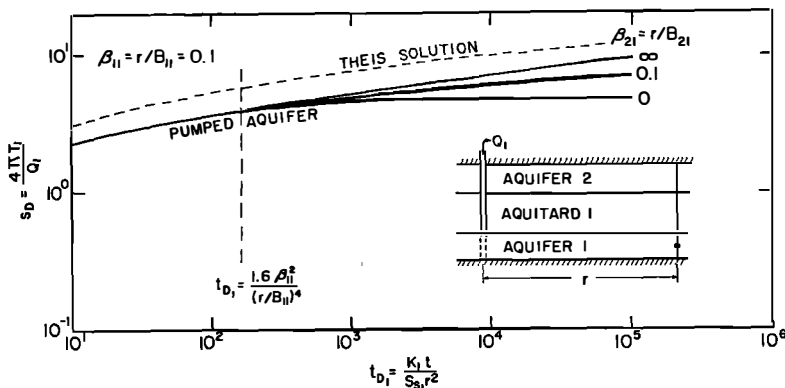


Fig. 8. Effect of unpumped aquifer on dimensionless drawdown in pumped aquifer.

$b_1'^2 e^{i\pi}]$ along CD . Defining the functions

$$F(y) = r^2 D[(y^2 \alpha_1' / b_1'^2) e^{i\pi}]$$

$$G(y) = \frac{A_2[(y^2 \alpha_1' / b_1'^2) e^{i\pi}] - \xi_1^2 [(y^2 \alpha_1' / b_1'^2) e^{i\pi}]}{D[(y^2 \alpha_1' / b_1'^2) e^{i\pi}]} - 1$$

which leads to equations 8 and 11, one can write

$$\lim_{\substack{\epsilon \rightarrow 0 \\ R \rightarrow \infty}} I_{CD} = \int_{-\infty}^0 (e^{-y^2 i D_1} - 1) \cdot [1 + G(y)] K_0(\xi_1^* r) \frac{dy}{y} \quad (\text{A15})$$

where $i D_1$ is defined by (7).

Similarly to evaluate $I_{D'C'}$, let $\lambda = (y^2 \alpha_1' / b_1'^2) e^{-i\pi}$ and $\xi_1^{**} = \xi_1[(y^2 \alpha_1' / b_1'^2) e^{-i\pi}]$ along $D'C'$, which leads to

$$\lim_{\substack{\epsilon \rightarrow 0 \\ R \rightarrow \infty}} I_{D'C'} = \int_0^{-\infty} (e^{-y^2 i D_1} - 1) \cdot [1 + G(y)] K_0(\xi_1^{**} r) \frac{dy}{y} \quad (\text{A16})$$

In considering the arguments of K_0 in (A15) and (A16) it is convenient to define

$$\omega_1^2(y) = \frac{(\xi_1^* r)^2}{e^{i\pi}} = \frac{(\xi_1^{**} r)^2}{e^{-i\pi}} \quad (\text{A17})$$

which leads to equation 9. Since $\omega_1^2(y)$ is real,

it follows that

$$\left. \begin{aligned} \arg(\xi_1^* r)^2 &= \pi \\ \arg(\xi_1^{**} r)^2 &= -\pi \end{aligned} \right\} \text{when } \omega_1^2(y) > 0 \quad (\text{A18})$$

$$\left. \begin{aligned} \arg(\xi_1^* r)^2 &= 0 \\ \arg(\xi_1^{**} r)^2 &= 0 \end{aligned} \right\} \text{when } \omega_1^2(y) < 0 \quad (\text{A19})$$

In addition since $I_0(v e^{\pm i\pi/2}) = J_0(v)$ for any v , then by virtue of (A17)

$$I_0(\xi_1^* r) = I_0(\xi_1^{**} r) = J_0[\omega_1(y)] \quad (\text{A20})$$

Based on equations A18, A19, and A20 and the expanded form of $K_0(\xi r)$ [McLachlan, 1961, p. 202], one can verify that

$$\begin{aligned} K_0(\xi_1^{**} r) - K_0(\xi_1^* r) &= \frac{1}{2} 1n(\xi_1^* r)^2 I_0(\xi_1^* r) - \frac{1}{2} 1n(\xi_1^{**} r)^2 I_0(\xi_1^{**} r) \\ &= \frac{i}{2} J_0[\omega_1(y)] [\arg(\xi_1^* r)^2 - \arg(\xi_1^{**} r)^2] \\ &= \begin{cases} i\pi J_0[\omega_1(y)] & \text{when } \omega_1^2(y) > 0 \\ 0 & \text{when } \omega_1^2(y) < 0 \end{cases} \quad (\text{A21}) \end{aligned}$$

Substituting (A15) and (A16) into equation A14 and using (A21), one can write

$$\begin{aligned} \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} (e^{\lambda t} - 1) \frac{A_2 - \xi_1^2}{D} K_0(\xi_1 r) \frac{d\lambda}{\lambda} \\ = \frac{1}{2} \int_0^\infty (1 - e^{-y^2 i D_1}) \cdot [1 + G(y)] J_0[\omega_1(y)] \frac{dy}{y} \quad (\text{A22}) \end{aligned}$$

where by virtue of (A21) the integral must be set to zero when $\omega_1^2(y) < 0$.

The second part of (A9) can be treated in a similar manner to obtain

$$\begin{aligned} \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} (e^{\lambda t} - 1) \frac{A_2 - \xi_2^2}{D} K_0(\xi_2 r) \frac{d\lambda}{\lambda} \\ = -\frac{1}{2} \int_0^\infty (1 - e^{-y^2 i D_1}) \end{aligned}$$

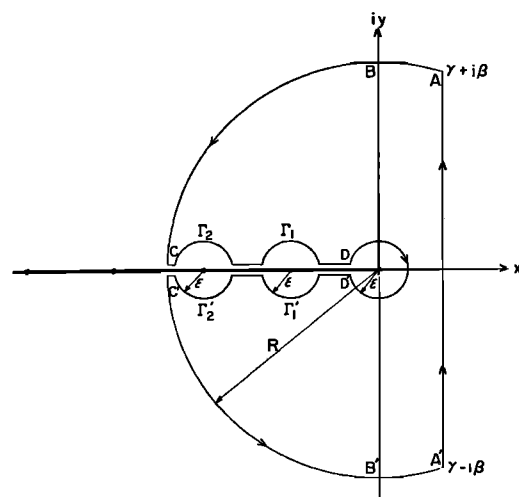


Fig. 9. Contour of integration used with Mellin's inversion formula.

$$[1 - G(y)]J_0[\omega_2(y)] \frac{dy}{y} \quad (A23)$$

where $\omega_2^2(y)$ is as defined in equation 10 and the integral must be set to zero when $\omega_2^2(y) < 0$.

Finally substituting (A9) into Mellin's inversion formula (A12) and using (A22) and (A23), one obtains the solution given by equation 4. The inversion of (A10) can be performed in essentially the same manner to obtain the solution given by equation 6.

To obtain the solution for the aquitard, tables of Laplace transforms [Spiegel, 1965, p. 252] can be used to obtain

$$L^{-1} \left[\frac{\sinh(\sqrt{p/\alpha_1'} z)}{\sinh(\sqrt{p/\alpha_1'} b_1')} \right] = \frac{2\pi\alpha_1'}{b_1'^2} \sum_{n=1}^{\infty} (-1)^n n \exp(-n^2 \pi^2 z_{D_1}) \sin \frac{n\pi z}{b_1'} \quad (A24)$$

The convolution of $(1 - e^{-v^* i_{D_1}})$ and $e^{-n^2 \pi^2 i_{D_1}}$ is $(1 - e^{-v^* i_{D_1}}) * e^{-n^2 \pi^2 i_{D_1}}$

$$= \frac{b_1'^2}{\alpha_1'} \left[\frac{1 - e^{-n^2 \pi^2 i_{D_1}}}{n^2 \pi^2} + \frac{e^{-n^2 \pi^2 i_{D_1}} - e^{-v^* i_{D_1}}}{n^2 \pi^2 - y^2} \right] \quad (A25)$$

With the aid of these two equations one can apply the convolution theorem [Carslaw and Jaeger, 1963, p. 7] to (A11) using the expressions for s_1 and s_2 as given by equations 4 and 6. Since $\sin[n\pi(1 - z/b_1')] = -(-1) \sin(n\pi z/b_1')$, the final solution can be written as given by equation 5.

APPENDIX B

To obtain an asymptotic solution in the aquitard at small values of time, we shall follow in part a procedure previously outlined by Hantush [1960]. Since t and p in the Laplace transform are inversely related, it is apparent that $b_1'^2 p/\alpha_1' \geq 10$ when $t \leq b_1'^2/10\alpha_1'$. In such case one can use the approximations $\coth(\sqrt{p/\alpha_1'} b_1') \approx 1$ and $\sqrt{p/\alpha_1'} b_1'/\sinh(\sqrt{p/\alpha_1'} b_1') \ll 1$ to obtain the following from (A5)

$$\left. \begin{aligned} A_1(p) &= \frac{p}{\alpha_1} + \frac{K_1' \sqrt{p/\alpha_1'}}{T_1} \\ B_1(p) &\ll A_1(p) \\ B_2(p) &\ll A_2(p) \end{aligned} \right\} \quad (B1)$$

Equations A3 and A4 therefore become

$$\left. \begin{aligned} \bar{s}_1 &= \frac{C}{a^2 + A_1} \\ \bar{s}_2 &= \frac{CB_2}{(a^2 + A_1)(a^2 + A_2)} \end{aligned} \right\} \quad (B2)$$

which shows that $\bar{s}_2 \ll \bar{s}_1$ and can be neglected in (A1) provided z does not approach b_1' too closely. As a result the inverse Hankel transform of (A1) can be obtained using (C4) as

$$\bar{s}_1' = \frac{Q_1}{2\pi T_1} \frac{\sinh[\sqrt{p/\alpha_1'}(b_1' - z)]}{\sinh(\sqrt{p/\alpha_1'} b_1')} \cdot \frac{K_0(r\sqrt{A_1})}{p} \quad (B3)$$

Using the approximation $\sinh(\sqrt{p/\alpha_1'} b_1') \approx \frac{1}{2} \exp(\sqrt{p/\alpha_1'} b_1')$ and the relationship

$$K_0(r\sqrt{A_1}) = \frac{1}{2} \int_0^{\infty} \exp\left[-y - \frac{r^2 A_1}{4y}\right] \frac{dy}{y}$$

Equation B3 can be rewritten

$$\bar{s}_1' = \frac{Q_1}{4\pi T_1} \int_0^{\infty} \frac{e^{-v}}{y} f(p) dy \quad (B4)$$

where

$$\begin{aligned} f(p) &= \frac{1}{p} \exp\left[-p \frac{r^2}{4\alpha_1 y}\right] \\ &\cdot \left\{ \exp\left[-\sqrt{p} \left(\frac{K_1' r^2}{4T_1 y \sqrt{\alpha_1'}} + \frac{z}{\sqrt{\alpha_1'}}\right)\right] \right. \\ &\quad \left. - \exp\left[-\sqrt{p} \left(\frac{K_1' r^2}{4T_1 y \sqrt{\alpha_1'}} + \frac{2b_1' - z}{\sqrt{\alpha_1'}}\right)\right] \right\} \end{aligned}$$

From tables of Laplace transforms we have

$$L^{-1}\left[\frac{1}{p} \exp(-c\sqrt{p})\right] = \operatorname{erfc}\left[\frac{c}{2\sqrt{t}}\right]$$

where $c > 0$, and this with theorem 5 of Carslaw and Jaeger [1963, p. 7] yields the inverse transform

$$L^{-1}[f(p)] = \begin{cases} 0 & \text{when } t \leq r^2/4\alpha_1 y \\ \operatorname{erfc}[g_1(y)] - \operatorname{erfc}[g_2(y)] & \text{when } t > r^2/4\alpha_1 y \end{cases}$$

where

$$g_1(y) = \frac{K_1' r^2 + 4T_1 z y}{8T_1 y \sqrt{\alpha_1'}(t - r^2/4\alpha_1 y)}$$

$$g_2(y) = \frac{K_1' r^2 + 4T_1(2b_1' - z)y}{8T_1 y \sqrt{\alpha_1'(t - r^2/4\alpha_1 y)}}$$

Using this result the inverse of (B4) can now be expressed in terms of dimensionless parameters as given in equation 22.

APPENDIX C

The infinite Hankel transform of order zero $F(a)$ of a function $f(r)$ is defined as

$$H[f(r)] = F(a) = \int_0^\infty r J_0(ar) f(r) dr \quad (C1)$$

The inverse of this transform is given by

$$H^{-1}[F(a)] = f(r) = \int_0^\infty a J_0(ra) F(a) da \quad (C2)$$

The particular relationships that are used in this work are

$$H\left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \bar{s}(r, p)\right)\right] = -a^2 \bar{s}(a, p) - \lim_{r \rightarrow 0} r \frac{\partial}{\partial r} \bar{s}(r, p) \quad (C3)$$

and

$$H^{-1}\left[\frac{a^{2n}}{a^2 + \xi^2}\right] = (-1)^n \xi^{2n} K_0(r\xi) \quad (C4)$$

where ξ is a real constant and n is a nonnegative integer. The details of the development have been shown elsewhere [Neuman and Witherspoon, 1969b].

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